

ANEENA MANOJ

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PROFESSIONAL EXPERIENCE

Orfiq

03/2025 – present | Frisco, United States

AI Engineer

- Leading GenAI development for Orfiq's learning platform by building LLM **Multi-Agents** with **LangChain**, **LangGraph**, **LangFlow**, and **Azure OpenAI** (GPT-4.1, Claude 3.5) to power course creation, tutoring, and personalized feedback across five core modules.
- Designed **Agentic RAG pipelines** with **AstraDB** to generate contextual responses from multi-modal inputs.
- Engineer and deploy secure, compliant microservices-based **RESTful** APIs using **FastAPI**, focusing on tasks like flashcard generation, copilot-style note summarization, and adaptive learning recommendations.
- Ensure comprehensive monitoring, observability, and robust error-handling frameworks using tools like **LangSmith** to maintain system reliability, security, and performance.
- Deploy solutions on **Azure Cloud** with stringent guardrails, ensuring compliance, security best practices, and scalability.

OpelSoft

02/2024 – 03/2025 | Cary, NC

AI Solutions Engineer

- Designed and implemented **Generative AI** solutions, integrating GPT-3.5 and LangChain to automate document processing, reducing manual effort by 30% and streamlining workflows.
- Integrated **Large Language Models (LLMs)** with **LangChain** to create advanced NLP pipelines, enabling insightful analysis of unstructured data and transforming decision-making processes.
- Implemented **AI agent** and multi-agent workflows using **Crew AI**, **LangGraph**, and **Smol Agents** enabling seamless orchestration between agents for information retrieval, and decision support.
- Built interactive web applications using **Streamlit** and **Flask**, enhancing claims officer productivity by 20% with AI-powered predictions and faster task completion.
- Deployed scalable solutions on **AWS** infrastructure (**S3**, **EC2**, **SageMaker**, **Bedrock**), achieving 99.9% uptime and supporting a 50% increase in concurrent users during peak operations.
- Optimized **SQL** and **MongoDB** queries, improving data retrieval performance by 50% and ensuring rapid access to critical claims data.
- Automated CI/CD pipelines with **Docker**, and **Kubernetes**, reducing deployment time by 40% and improving system reliability during releases.

IQVIA

07/2021 – 12/2021 | Bangalore, India

Software Developer

- Designed and implemented **REST APIs** using **Python**, ensuring seamless communication between system components.
- Utilized **Postman** for comprehensive endpoint testing, guaranteeing robust functionality and reliability.
- Spearheaded data preprocessing pipelines with **Pandas** for large datasets, improving model training efficiency by 30%.
- Architected and optimized **PostgreSQL** schema, enhancing data retrieval performance by 20% and ensuring scalable management.
- Directed **data augmentation** techniques to expand training datasets, boosting chatbot generalization accuracy by 15%.

Hexaware Technologies

10/2020 – 06/2021 | Chennai, India

Associate Software Developer

- Spearheaded summarization tools using **spaCy** and **BERT** models for enterprise email categorization, processing over 2 million emails monthly.

- Developed **Python**-based APIs with **Flask** to automate technical documentation retrieval, reducing manual effort and ensuring up-to-date information delivery.
- Deployed **Azure Cognitive Services** for multilingual text analytics solutions, enabling feedback processing in 12+ languages for global teams.
- Designed **SQL Server** solutions for data warehousing, consolidating logs across distributed systems and improving analytics query execution times.

Cochin University of Science and Technology

04/2019 – 05/2019 | Kochi, India

Project Manager Intern

- Directed a team of 5 to develop a comprehensive hostel management website for over 3,000 students with a responsive UI, reducing manual processes by 80% and improving user interaction.
- Built a robust web application using **Python**, **CSS**, and the **Django** framework, integrating analytics dashboards for tracking occupancy, ticket management, and user interactions.
- Implemented **RESTful APIs** with Django REST Framework for seamless data flow and real-time ticket resolution.
- Designed and optimized relational database schemas with **SQL**, enhancing query performance by 40% and ensuring rapid data access.
- Utilized **NLP** techniques for analyzing feedback and support tickets, automating text categorization to improve issue resolution.
- Automated reporting with Python libraries - **Matplotlib**, and **Seaborn**, reducing manual effort by 40% and generating actionable insights.

TECHNICAL SKILLS

Programming Languages – Python, Java, C++, C#, JavaScript

Data Engineering – ETL Pipelines, Data Modeling, Data Warehousing, PyTorch, Pandas, Spark

Generative AI – LangChain, LangFlow, Transformers, Scikit Learn, RAGStack, Crew AI, LangGraph, AI agent

Database Management – SQL, MySQL, Oracle, MongoDB, CHROMA, AstraDB, Pinecone

Big Data & Cloud – AWS (Redshift, S3, Glue, EMR, Lambda, IAM), Azure OpenAI, Google Cloud Platform (GCP), Vertex AI,

Web Development and APIs – FastAPI, REST APIs, Microservices, HTML, CSS

DevOps and LLM OPs – Docker, Kubernetes, Git, Streamlit, Langfuse, Langsmith

PROJECTS

BrandCraft Toolkit

LangChain, LangSmith, LangFlow, Python, AstraDB, FastAPI, Google Cloud, Prompt Engineering

- Developed a marketing AI toolkit using **LangChain** to automate public relations, ensure consistent brand quality, and reduce manual effort by 90%.
- Leveraged **LangFlow** for prototyping workflows and **LangSmith** for observability, enabling real-time monitoring and optimization of AI processes.
- Implemented a dynamic **LLM model-switching** mechanism to adapt AI outputs to diverse marketing scenarios.
- Deployed the solution on **Google Cloud** with hybrid search and **Retrieval-Augmented Generation (RAG)** for scalable and efficient content creation.

Models: Vertex AI Gemini, Anthropic Claude, OpenAI GPT-4.0, OpenAI GPT-4.1 Preview, GCP

Multimodal Fashion AI: Powered by Astra DB & Vertex AI

DataStax Astra DB, Langchain, Vertex AI, Streamlit

- Developed a **Multimodal** Fashion AI app with **Astra DB** and **Vertex AI**, utilizing **LangChain** via DataStax **RAGStack** for precise fashion item matching from photos, enhancing user experience.
- Deployed the app on **Streamlit**, utilizing advanced AI techniques for real-time, accurate apparel matching and performance optimization, improving system efficiency.

EDUCATION

Master of Science, Computer Science <i>The University of Texas at Dallas, CGPA : 3.82/4</i>	01/2022 – 12/2023 Richardson, Texas
Bachelor of Technology, Computer Science and Engineering <i>Cochin University of Science and Technology, CGPA: 8.14/10</i>	08/2016 – 09/2020 Kochi, India

AWARDS

Best new joiner award: Hexaware Technologies	2020
Best Inclusive Product : Make-A-Ton Hackathon <i>Amazon Go: Smart Shopping</i>	2019